

HYPOTHESIS TESTING

P-values, Type II error, and power

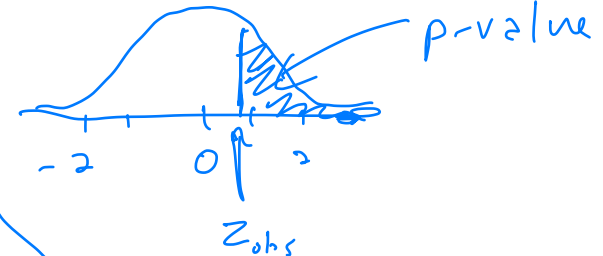
p -values

- An alternative to determining a rejection region and making a decision based on that is to calculate a p -value and compare that with the (pre-selected) α .
- The p -value is defined to be the probability, given that H_0 is true, that the test statistic would be at least as extreme as what was actually observed.
- For the normally (a.i.a.) distributed test statistics, the p -value depends on the alternative hypothesis.

• $H_a: \theta > \theta_0$ ("upper tailed")

$$p\text{-value} = P(Z > z_{obs})$$

observed value of T.S.

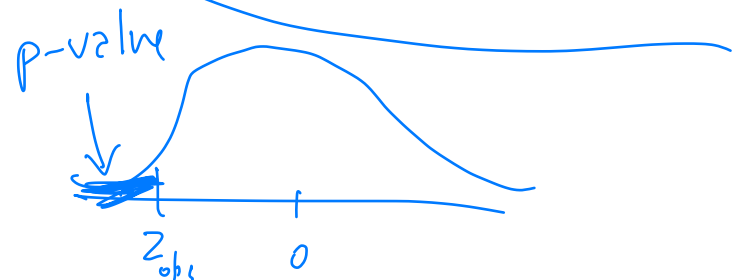
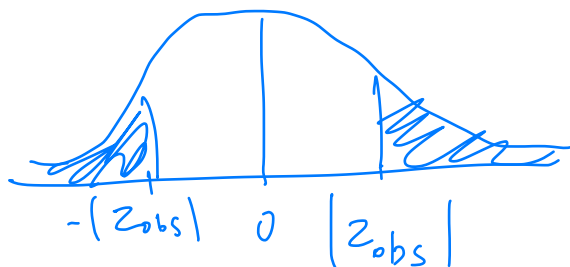


• $H_a: \theta < \theta_0$ ("lower tailed")

$$P(Z < z_{obs})$$

• $H_a: \theta \neq \theta_0$ ("two tailed")

$$p\text{-value} = P(Z > |z_{obs}|)$$



$H_0: \theta = \theta_0$

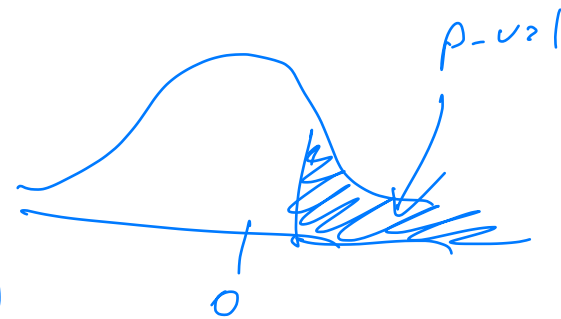
Examples

- Calculate the p -value for Examples 3 and 4.

③ $Z_{obs} = 0.652$

$H_a \rightarrow$

$$p\text{-value} = P(Z > 0.652)$$



$$1 - \text{pnorm}(0.652) = \boxed{0.257}$$

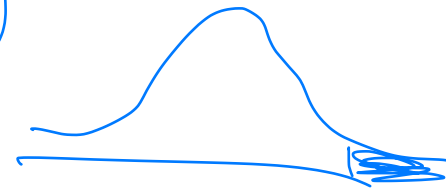
④ $Z_{obs} = 1.596$

$H_a \rightarrow$

$$p\text{-value} = P(Z > 1.596)$$

$$1 - \text{pnorm}(1.596)$$

$$= \boxed{0.0552}$$



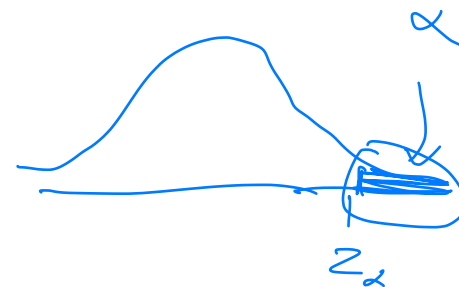
Calculating Type II error probability for "Z test"

- If $Y_1, Y_2, \dots, Y_n \sim iid N(\mu, \sigma^2)$ (suppose σ^2 is "known") then a test for $H_0: \mu = \mu_0$ will depend on a "Z test statistic":

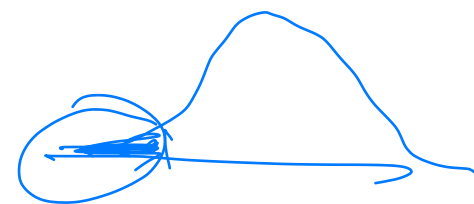
$$Z = \frac{\bar{Y} - \mu_0}{\sigma / \sqrt{n}}$$

The "rejection region" for this test depends on H_a .

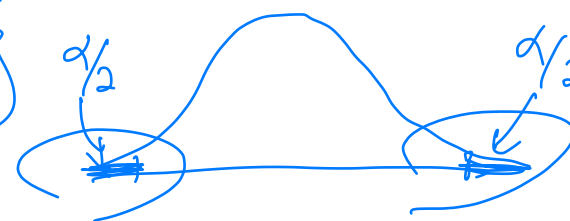
$$H_a: \mu > \mu_0 \quad RR \{Z: Z > z_\alpha\}$$



$$H_a: \mu < \mu_0 \quad RR \{Z: Z < -z_\alpha\}$$



$$H_a: \mu \neq \mu_0 \quad RR \{Z: |Z| > z_{\alpha/2}\}$$



$$RR \text{ for } \bar{Y} \\ \{ \bar{Y} : \bar{Y} < 10.684 \}$$

Calculating Type II error probability

$$\sigma^2 = 16 \quad \sigma = 4$$

- Example 10: $Y_1, Y_2, \dots, Y_n \sim iid N(\mu, 16)$ and the hypotheses are $H_0: \mu = 12$ vs $H_a: \mu < 12$. If $\alpha = 0.05$ and $n = 25$, calculate the Type II error probability if the true mean is 10.

$$RR = \{ Z : Z < -1.645 \}$$

$$TS \quad Z = \frac{\bar{Y} - 12}{4/5} \quad \text{observed}$$

$$\text{Reject } H_0 \text{ if } Z \in RR \\ \text{if } Z < -1.645$$

$$\text{if } \frac{\bar{Y} - 12}{4/5} < -1.645$$

$$\text{if } \bar{Y} < 12 - \frac{4}{5}(1.645) = \underline{10.684}$$

equivalent

$$\beta = P(\text{Type II error}) = P(\text{don't reject } H_0)$$

$$= P(Z \notin RR_Z)$$

$$= P(\bar{Y} \notin RR_{\bar{Y}})$$

$$= P(\bar{Y} \geq 10.684)$$

$$= P\left(\frac{\bar{Y} - 10}{4/5} \geq \frac{10.684 - 10}{4/5}\right)$$

$$= P(Z \geq 0.855)$$

$$1 - \text{pnorm}(0.855)$$

$$\underline{0.196}$$

Assuming

$\mu = 10$

$\Downarrow$

$\bar{Y} \sim N(10, \frac{16}{25})$



$$\begin{aligned} \bar{X} - \bar{Y} &\sim N(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}) \\ \hline \bar{X} - \bar{Y} &\sim N(\mu_1 - \mu_2, \frac{20}{n}) \end{aligned}$$

Calculating Type II error probability

- Example 11: $X_1, X_2, \dots, X_n \sim iid N(\mu_1, 10)$ and $Y_1, Y_2, \dots, Y_n \sim iid N(\mu_2, 10)$ and the hypotheses are $H_0: \mu_1 - \mu_2 = 0$ vs $H_a: \mu_1 - \mu_2 \neq 0$. If $\alpha = 0.01$ and $n = 49$, calculate the Type II error probability if $\mu_1 = 15$ and $\mu_2 = 17$.

$$T.S. \quad \frac{\bar{X} - \bar{Y} - 0}{\sqrt{\frac{20}{n}}} = Z$$

$$RR: \{Z : |Z| > z_{\alpha/2}\}$$

$$\text{Reject } H_0 \text{ if } \left| \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{20}{n}}} \right| > z_{\alpha/2}$$

$$\text{"Accept } H_0" \text{ if } \left| \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{20}{n}}} \right| < z_{\alpha/2}$$

$$\left[-z_{\alpha/2} < \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_0^2}{n}}} < z_{\alpha/2} \right] \quad \text{"Accept } H_0 \text{"}$$

$$\frac{\bar{X} - \bar{Y} - 2}{\sqrt{\frac{\sigma_0^2}{n}}}$$

P(Type II error) if $\mu_1 = 15, \mu_2 = 17$?

P("accept H_0 ") " " "

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_0^2}{n}}} < z_{\alpha/2}\right)$$

If $\mu_1 = 15, \mu_2 = 17$
then this does not
have a $N(0,1)$ dist.

$$P\left(-z_{\alpha/2} - \frac{2}{\sqrt{\frac{\sigma_0^2}{n}}} < \boxed{\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_0^2}{n}}} - \frac{2}{\sqrt{\frac{\sigma_0^2}{n}}}} < z_{\alpha/2} - \frac{2}{\sqrt{\frac{\sigma_0^2}{n}}}\right)$$

$\sim N(0,1)$

$$\alpha = 0.01$$

$$z_{.005} = 2.576$$

$$P\left(-z_{\alpha/2} - \frac{2}{\sqrt{\frac{\sigma_0^2}{n}}} < Z < z_{\alpha/2} - \frac{2}{\sqrt{\frac{\sigma_0^2}{n}}}\right) = P\left(-2.576 - \frac{2}{\sqrt{\frac{20}{49}}} < Z < 2.576 - \frac{2}{\sqrt{\frac{20}{49}}}\right)$$

$$pnorm(2.576 - 2/\sqrt{20/49}) - pnorm(-2.576 - 2/\sqrt{20/49})$$

$$0.28962 = P(\text{Type II error})$$

$$\begin{aligned} \text{Power} &= P(\text{Correctly reject } H_0) \\ &= 1 - P(\text{Type II error}) \end{aligned}$$

$\mu_1 = 15$ $\mu_2 = 17$ power = $1 - 0.28962$
 $\searrow$ $\swarrow$ = 0.71038

power depends on
true θ

Power

- Definition 2: The POWER of a hypothesis test for parameter value θ is the probability of rejecting H_0 if the true parameter value is equal to θ .
- For any θ , $\text{power}(\theta) = 1 - P(\text{type II error}) = 1 - \beta$, where the probability is computed assuming that the true value of the parameter is θ .
- There is a tradeoff between α and β !

